OpenChuckCAM user guide

OpenChuckCAM turns a 3D model into G-code for 4-axis rotary machining: the work is held at one end in a chuck and turned about its length while the cutter moves along and around it. The first machine it supports is the Makera Z1.

This guide is for people who want to make parts with the program, including people who have not used CAM software before. It explains each term the first time it is used and collects them in the glossary.

> **WARNING**
>
> Generated G-code must be checked by you before it is run on a machine. OpenChuckCAM does not guarantee that a program is safe or correct for any machine, tool or setup. The program is in early development, and very little of its output has been run on a real machine. Read *Checking G-code before you run it* before you cut anything.

Where to start

New to the program
: Read *Install and first run*, then work through Your first project. The tutorial takes one small part from a model file to G-code in ten steps, with a screenshot of each.

New to CAM or to 4-axis work
: The *Concepts* pages explain what a rotary axis does, what stock and a held region are, and why a job is usually cut in two passes, roughing and then finishing.

Ready to run a program on the Makera Z1
: Follow *Air cut on the Makera Z1* and the *Safe-practice checklist*. An air cut runs the program with no tool and no material, so you can watch the moves before anything can be damaged.

Looking for a setting or a word
: Use the search box, the glossary, or the *Reference* pages.

What the program does

Step in the program	What you do there
Model	Import an STL or STEP file, check it, and place it on the rotary axis.
Setup	Choose the machine, describe the stock and how the work is held.
Operations	Choose tools and add the cutting operations, from roughing to finishing.
Simulate	Generate the toolpaths, watch the cut, and compare the result with the model. Then post and export the G-code and a setup sheet.

About this guide

This guide describes the development version of OpenChuckCAM. Each release has its own copy; use **Version** at the top of the page to switch. The guide is part of the project's source and is licensed AGPL-3.0-or-later. To report a mistake in it, open an issue in the project's repository.

Your first project

This tutorial takes you through one complete job, from an empty window to G-code and a setup sheet. The part is a small turned and fluted piece, a baluster with a ball finial, rough-cut with one tool and finished with another, for a Makera Z1 with a chuck.

> **NOTE**
> This page is a prototype. Only step 5 is written here; the full tutorial is seeded from lane DOCSUSER's text.

5. Hold the work

The chuck grips one end of the stock. In this step you tell the program how much of that end is gripped, so that no toolpath goes there.

> **NOTE**
>
> The screenshots in this tutorial were taken from an earlier build, before the workflow changed from five steps to four. The settings shown are the same; the step names along the top of the window differ.

Set the held length

1. In the Inspector, open **Workholding and tabs**.
2. Tick **Held at the start**.
3. Set **Grip length** to `8` . Lengths are in millimetres.
4. Press **Apply** at the bottom of the section.

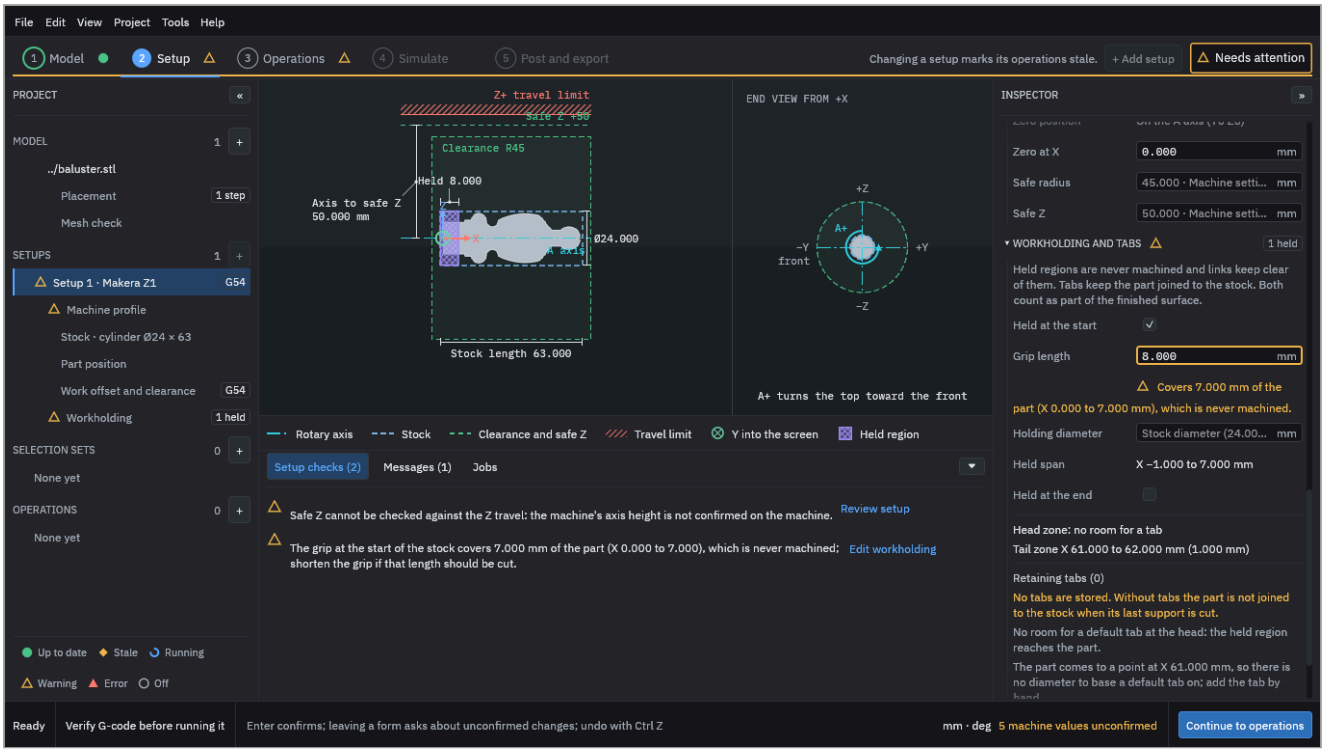


Figure 5. An 8 mm held region at the start of the stock, drawn hatched in the view.

The first 8 mm of the stock, from X = −1 to X = 7, is now a held region. It is never cut, and the toolpaths keep clear of it.

The program warns that this length covers 7 mm of the part, the plain end, which therefore stays as bar. That is what you want here. The warning matters when the held length covers part of the shape you meant to cut.

Tabs

The program also reminds you that, without tabs, the part is not joined to the stock once its last support is cut. This piece stays joined to the held bar, so the tutorial uses no tab. If your own job cuts the part free, add tabs.

Holding the far end

If you also support the far end of the stock, for example with a tailstock, tick **Held at the end** in the same way.

> **WARNING**
>
> A held region is a length along the axis, not a 3D body. OpenChuckCAM does not model the chuck, its jaws or a tailstock, and does not check them against the tool or the holder. Check these clearances yourself, on the machine, before you run a program.

When the held length is set, press **Continue to operations**.

Glossary

The words this guide uses for machining and for the program. Each page links a term here the first time it uses it.

A · C · G · H · P · R · S · T

A axis

The rotary axis that turns the work. On the Makera Z1 it turns about the X axis, the line along the length of the work. Positive A is a turn in one fixed direction; the machine profile records which. See also rotary axis.

CAM

Computer-aided manufacturing: software that works out the moves a machine tool must make to cut a part from a model, and writes them as G-code.

G-code

The text program a CNC machine's controller reads: one move or setting per line, such as `G1 X10 A90 F600`. Each controller reads its own variant, which is why the program needs a post-processor.

Held region

A length at one end of the stock, measured along the axis, that the chuck (or a tailstock) grips. It is never cut, and toolpaths keep clear of it. The program knows it only as a length, not as a solid body, so it cannot check the chuck itself against the tool.

Post-processor

The part of the program that turns toolpaths into the G-code variant a particular controller reads. Often shortened to *post*. OpenChuckCAM includes a post for the Makera Z1 and one for LinuxCNC.

Rotary axis

An axis that turns instead of moving in a straight line. In 4-axis rotary work the rotary axis holds the work and turns it, so the cutter can reach all the way round. See also A axis.

Stepover

How far the cutter moves sideways between neighbouring passes. A smaller stepover leaves a smoother surface and takes longer.

Stock

The material you start with, such as a round bar or a block. The program cuts the part out of the stock, and needs its shape and size to know where material is.

Tab

A small bridge of material left uncut so that the part stays attached to the stock until you separate it yourself.

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